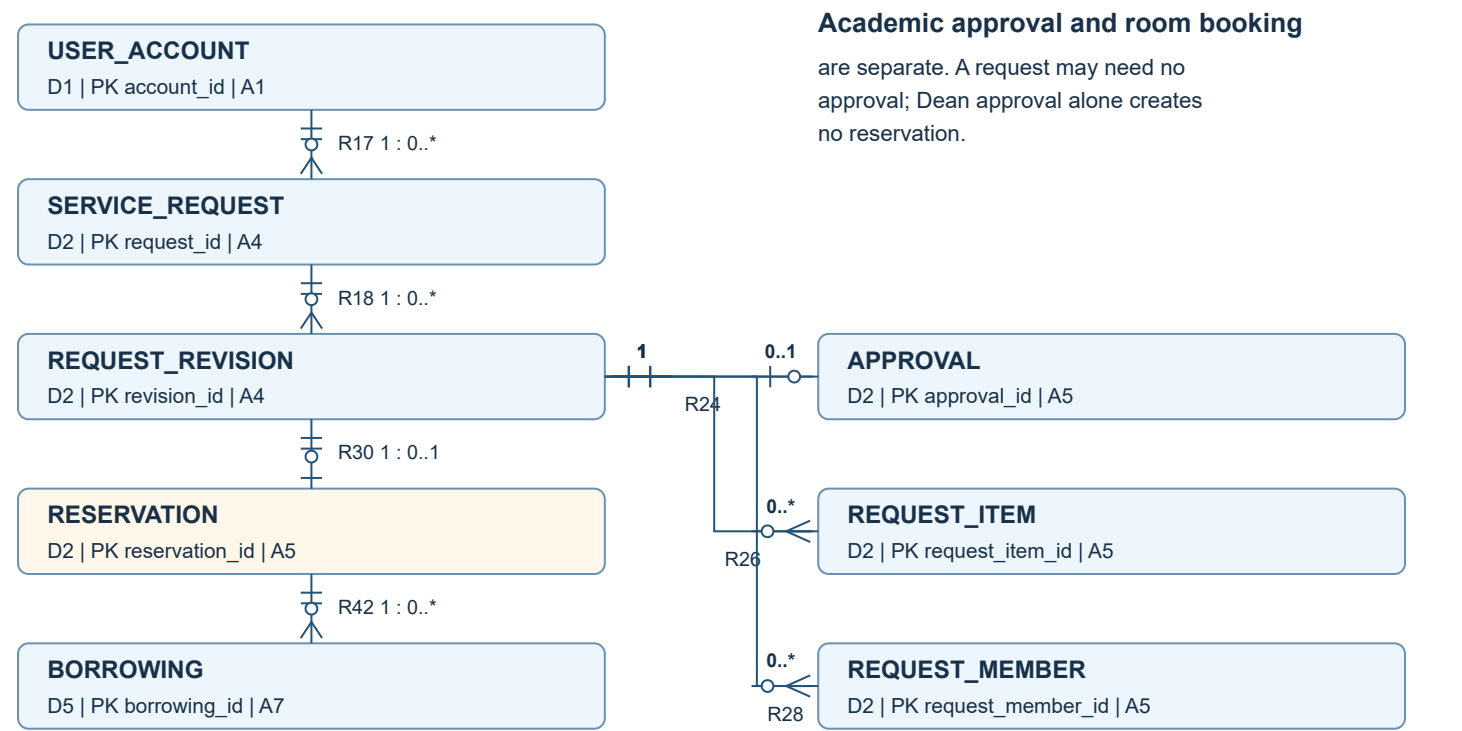


NU FAIRVIEW LABORATORY MANAGEMENT SYSTEM

Connected ERD - schema overview

30 entities | 208 attributes | 61 internal PK-FK relationships



A1 - Accounts and Student Identity

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

USER_ACCOUNT				D1 Demo
PK	account_id	uuid		
FK/U	student_id ?	uuid	R01	
FK	staff_lab_id ?	uuid	R02	
UK	login_identifier	text	UK	
UK	email	text	UK	
	first_name	text		
	middle_name ?	text		
	last_name	text		
	contact_number ?	text		
	department ?	text		
	role	enum		
	status	enum		
	notes ?	text		
	created_at	timestampz		
	updated_at	timestampz		

Profile and role context for login, account management and staff scoping. account_id maps to Supabase auth.users.id at integration.

Passwords belong to Supabase Auth; no password_hash or temporary password is stored in the profile.

STUDENT				D1 Demo
PK	student_id	uuid		
UK	student_number	text	UK	
	full_name	text		
	program ?	text		

Participant identity; NU ID is independent of the submitting Class Representative account.

student_number matches YYYY-NNNNNNNN, e.g. 2024-1031816.

SECTION				D1 Demo
PK	section_id	uuid		
UK	section_code	text	UK	
	program ?	text		
	active	boolean		

Named class section used by Faculty assignments and the fixed participant section.

A2 - Faculty, Classrep and Class Membership

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

FACULTY_SECTION				D1 Demo
PK	faculty_section_id	uuid		
FK	faculty_id	uuid	R03	
FK	section_id	uuid	R04	
FK	lab_id	uuid	R05	
	active	boolean		

Many assigned sections per Faculty, scoped to a laboratory before a subject/class is created.

Unique: faculty_id + section_id + lab_id.

CLASS_REP_ASSIGNMENT				D1 Demo
PK	rep_assignment_id	uuid		
FK	account_id	uuid	R08	
FK	section_id	uuid	R09	
FK	group_id	uuid	R10	
	active	boolean		
	assigned_at	timestampz		

Headlab Classrep form: fixed section, designated Faculty, laboratory and existing subject/class.

account_id must reference a CLASSREP account.

CLASS_GROUP				D1 Demo
PK	group_id	uuid		
FK	term_id	uuid	R06	
FK	faculty_section_id	uuid	R07	
	subject_code	text		
	subject_name	text		
	active	boolean		

Subject/class assignment. This table does not mean the Group request type.

Unique: term_id + faculty_section_id + subject_code.

GROUP_MEMBER				D1 Future
PK	group_member_id	uuid		
FK	group_id	uuid	R11	
FK	student_id	uuid	R12	
	active	boolean		

Authorized roster for the assigned subject/class; required for real participant ownership checks.

Unique: group_id + student_id.

A3 - Laboratories, Rooms and Regular Schedule

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

LABORATORY				D3 Demo
PK	lab_id	uuid		
UK	lab_code	text	UK	
	lab_name	text		
	active	boolean		

Physics or Circuits laboratory scope; one laboratory can contain several rooms.

lab_code is PHYSICS or CIRCUITS. Staff data access uses this ID, not a room ID.

LAB_ROOM				D3 Demo
PK	room_id	uuid		
FK	lab_id	uuid	R13	
	room_name	text		
	active	boolean		

Room Number choices under a laboratory, e.g. Physics 201/202 and Circuits 301/302.

Unique: lab_id + room_name.

REGULAR_SCHEDULE				D3 Demo
PK	regular_schedule_id	uuid		
FK	term_id	uuid	R14	
FK	room_id	uuid	R15	
FK	group_id ?	uuid	R16	
	block_type	enum		
	weekday	integer		
	start_time	time		
	end_time	time		
	valid_from	date		
	valid_until	date		

Recurring laboratory or lecture blocks rendered by Room Availability.

weekday is 1-6 (Monday-Saturday) in the current form.
start_time < end_time.

TERM				D3 Future
PK	term_id	uuid		
UK	term_code	text	UK	
	starts_on	date		
	ends_on	date		

Term boundary for subject assignments and weekly schedules.

starts_on <= ends_on.

A4 - Service Requests and Immutable Revisions

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

SERVICE_REQUEST				D2 Demo
PK	request_id	uuid		
UK	reference_no	text	UK	
FK	requester_id	uuid	R17	
	status	enum		
	submitted_at	timestampz		

Submitted request header shown in Dean, requester status and laboratory Staff request queues.

requester_id must reference CLASSREP or FACULTY. Identity, reference and initial status are derived on the server.

REQUEST_REVISION					D2 Demo
PK	revision_id	uuid			
FK	request_id	uuid	R18		
FK	lab_id	uuid	R19		
FK	group_id	uuid	R20		
FK	room_id	uuid	R21		
FK	regular_schedule_id ?	uuid	R22		
FK	created_by	uuid	R23		
	revision_no	integer			
	request_type ?	enum			
	activity_type ?	enum			
	schedule_type	enum			
	request_for	enum			
	starts_at	timestampz			
	ends_at	timestampz			
	notes ?	text			
	is_current	boolean			
	created_at	timestampz			

Detached request snapshot: laboratory, class, room/time, request/activity type and notes.

Unique: request_id + revision_no.

A5 - Approval, Requested Items and Participants

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

APPROVAL			D2 Demo
PK	approval_id	uuid	
FK/U	revision_id	uuid	R24
FK	approver_id	uuid	R25
	reviewer_role	enum	
	decision	enum	
	remarks ?	text	
	created_at	timestamptz	
	decided_at ?	timestamptz	

Exactly one selected Faculty or Dean reviewer where academic approval is required.

Classrep On-Schedule -> assigned Faculty. Classrep Out-of-Schedule -> selected assigned Faculty or Dean.

RESERVATION			D2 Future
PK	reservation_id	uuid	
FK/U	revision_id	uuid	R30
FK	processed_by	uuid	R31
	status	enum	
	reserved_at	timestamptz	

Confirmed dated room booking, created only by later laboratory processing after any required academic approval.

No reservation is created by the current Submit or Dean Approve actions.

REQUEST_ITEM			D2 Demo
PK	request_item_id	uuid	
FK	revision_id	uuid	R26
FK	item_id ?	uuid	R27
	name_snapshot	text	
	kind_snapshot	enum	
	unit_snapshot	text	
	qty_requested	integer	

Requested catalogue items or additional manual equipment/materials; quantities are not issued stock.

item_id may be null for Additional Items; name/type/quantity are still mandatory.

REQUEST_MEMBER			D2 Demo
PK	request_member_id	uuid	
FK	revision_id	uuid	R28
FK	student_id	uuid	R29
	name_snapshot	text	
	student_no_snapshot	text	

Selected students for Classrep Group or Student Only, on both schedule variants.

Unique: revision_id + student_id.

A6 - Laboratory Inventory and Stock Ledger

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

ITEM_CATEGORY				D4 Demo
PK	category_id	uuid		
UK	category_name	text	UK	

Category tags for equipment/material search and inventory management.

STOCK_MOVEMENT				D4 Future
PK	movement_id	uuid		
FK	item_id	uuid	R34	
FK	recorded_by	uuid	R35	
FK	borrowing_item_id ?	uuid	R36	
FK	return_entry_id ?	uuid	R37	
FK	disposal_id ?	uuid	R38	
	on_hand_delta	integer		
	owned_delta	integer		
	movement_type	enum		
	recorded_at	timestampz		

Auditable stock ledger for receipts, adjustments, issues, returns, consumption and disposal.

Update ITEM stock and the source transaction in the same database transaction. Ledger posting is not yet implemented.

ITEM				D4 Demo
PK	item_id	uuid		
FK	lab_id	uuid	R32	
FK	category_id	uuid	R33	
	item_name	text		
	item_type	enum		
	unit	text		
	stock_quantity	integer		
	condition_status	enum		
	reorder_level	integer		
	active	boolean		
	updated_at	timestampz		

Inventory row owned by one laboratory; shared equipment types still have distinct stock per lab.

Unique: lab_id + item_name.

DISPOSAL				D10 Future
PK	disposal_id	uuid		
FK	item_id	uuid	R39	
FK	recorded_by	uuid	R40	
FK	return_entry_id ?	uuid	R41	
	quantity	integer		
	waste_class	text		
	reason	text		
	disposed_at	timestampz		
	status	enum		

Documented physical waste disposal; lost items are not physical waste.

Disposal quantity must be positive and supported by physical stock or a damaged return.

A7 - Borrowing Slips, Issuance and Returns

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

BORROWING			D5 Demo
PK	borrowing_id	uuid	
FK	reservation_id	uuid	R42
FK	borrower_account_id	uuid	R43
FK	issued_by	uuid	R44
UK	slip_no	text	UK
	issued_at	timestampz	
	due_at	timestampz	
	status	enum	
	closed_at ?	timestampz	

Borrowing slip header for the Staff record screen; issuance is a future transaction.

Current Staff borrowing slips are isolated sample records, not created from demo requests. The target FK joins a slip to a confirmed reservation.

RETURN_ENTRY			D5 Future
PK	return_entry_id	uuid	
FK	borrowing_item_id	uuid	R50
FK	received_by	uuid	R51
	qty_good	integer	
	qty_broken	integer	
	qty_lost	integer	
	qty_consumed	integer	
	recorded_at	timestampz	

Partial-return history and outcome quantities for each issued item.

Quantities are nonnegative and total reconciled quantity cannot exceed issued quantity. At least one outcome quantity is positive.

BORROWING_ITEM			D5 Future
PK	borrowing_item_id	uuid	
FK	borrowing_id	uuid	R45
FK	request_item_id	uuid	R46
FK	item_id	uuid	R47
	qty_issued	integer	

Actual issued quantities, separate from requested quantities.

Unique: borrowing_id + request_item_id + item_id.

BORROWING_MEMBER			D5 Future
PK	borrow_member_id	uuid	
FK	borrowing_id	uuid	R48
FK	request_member_id	uuid	R49

Accountable selected students, never all students in a section automatically.

Unique: borrowing_id + request_member_id.

A8 - Student Clearance and Laboratory Logs

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

CLEARANCE			D6 Future
PK	clearance_id	uuid	
FK	return_entry_id	uuid	R52
FK	borrow_member_id	uuid	R53
FK	raised_by	uuid	R54
FK	settled_by ?	uuid	R55
	quantity	integer	
	reason	text	
	status	enum	
	raised_at	timestampz	
	settled_at ?	timestampz	

Student liabilities from a documented return and accountable selected participant.

Clearance remains a Classrep/student workflow. Faculty has no clearance page or implied student participants.

USAGE_LOG			D11 Future
PK	usage_id	uuid	
FK	reservation_id	uuid	R56
FK	borrowing_id ?	uuid	R57
FK	recorded_by	uuid	R58
	actual_start	timestampz	
	actual_end	timestampz	
	remarks ?	text	

Actual laboratory use after a reservation; distinct from planned time.

Room-only use can have no borrowing slip. Faculty identity comes from the reservation class assignment.

DAILY_TASK			D7 Future
PK	task_id	uuid	
FK	lab_id	uuid	R59
FK	recorded_by	uuid	R60
	task_date	date	
	activity	text	
	status	enum	
	work_availability	text	
	overtime_hours	numeric	
	remarks ?	text	

Laboratory Staff work log retained for the broader documented system.

A9 - Lab Assistant Extension

Full definitions | ? = nullable | PK = primary key | FK = foreign key | U / UK = unique | Rxx = relationship ID

KNOWLEDGE_ARTICLE			D8 Future
PK	article_id	uuid	
	topic	text	
	content	text	
	category	text	
	updated_at	timestampz	

Curated information for the future Lab Assistant; not an approval engine.

CHAT_EXCHANGE			D9 Future
PK	chat_id	uuid	
FK	requester_id	uuid	R61
	question	text	
	answer	text	
	intent	enum	
	response_status	enum	
	asked_at	timestampz	

Future authenticated Lab Assistant question/answer history.

Chat does not approve requests, reserve a room, mutate inventory or create a Forecast entity.

Cross-table lifecycle/role rules are listed in the dictionary and the final implementation boundary page.

PK-FK relationship register 1 / 4

Left cardinality = parents per child. Right = children per parent. A1-A9 identify full-definition pages.

ID	Parent / PK	Cardinality	Child / FK	Pages
R01	STUDENT student_id	0..1 — 0..1	USER_ACCOUNT student_id	A1 A1
R02	LABORATORY lab_id	0..1 — 0..*	USER_ACCOUNT staff_lab_id	A3 A1
R03	USER_ACCOUNT account_id	1 — 0..*	FACULTY_SECTION faculty_id	A1 A2
R04	SECTION section_id	1 — 0..*	FACULTY_SECTION section_id	A1 A2
R05	LABORATORY lab_id	1 — 0..*	FACULTY_SECTION lab_id	A3 A2
R06	TERM term_id	1 — 0..*	CLASS_GROUP term_id	A3 A2
R07	FACULTY_SECTION faculty_section_id	1 — 0..*	CLASS_GROUP faculty_section_id	A2 A2
R08	USER_ACCOUNT account_id	1 — 0..*	CLASS_REP_ASSIGNMENT account_id	A1 A2
R09	SECTION section_id	1 — 0..*	CLASS_REP_ASSIGNMENT section_id	A1 A2
R10	CLASS_GROUP group_id	1 — 0..*	CLASS_REP_ASSIGNMENT group_id	A2 A2
R11	CLASS_GROUP group_id	1 — 0..*	GROUP_MEMBER group_id	A2 A2
R12	STUDENT student_id	1 — 0..*	GROUP_MEMBER student_id	A1 A2
R13	LABORATORY lab_id	1 — 0..*	LAB_ROOM lab_id	A3 A3
R14	TERM term_id	1 — 0..*	REGULAR_SCHEDULE term_id	A3 A3
R15	LAB_ROOM room_id	1 — 0..*	REGULAR_SCHEDULE room_id	A3 A3
R16	CLASS_GROUP group_id	0..1 — 0..*	REGULAR_SCHEDULE group_id	A2 A3

0..* means zero or many. A saved request needs one current revision; FK notation alone cannot enforce that.

PK-FK relationship register 2 / 4

Left cardinality = parents per child. Right = children per parent. A1-A9 identify full-definition pages.

ID	Parent / PK	Cardinality	Child / FK	Pages
R17	USER_ACCOUNT account_id	1	SERVICE_REQUEST requester_id	A1
				A4
R18	SERVICE_REQUEST request_id	1	REQUEST_REVISION request_id	A4
				A4
R19	LABORATORY lab_id	1	REQUEST_REVISION lab_id	A3
				A4
R20	CLASS_GROUP group_id	1	REQUEST_REVISION group_id	A2
				A4
R21	LAB_ROOM room_id	1	REQUEST_REVISION room_id	A3
				A4
R22	REGULAR_SCHEDULE regular_schedule_id	0..1	REQUEST_REVISION regular_schedule_id	A3
				A4
R23	USER_ACCOUNT account_id	1	REQUEST_REVISION created_by	A1
				A4
R24	REQUEST_REVISION revision_id	1	APPROVAL revision_id	A4
				A5
R25	USER_ACCOUNT account_id	1	APPROVAL approver_id	A1
				A5
R26	REQUEST_REVISION revision_id	1	REQUEST_ITEM revision_id	A4
				A5
R27	ITEM item_id	0..1	REQUEST_ITEM item_id	A6
				A5
R28	REQUEST_REVISION revision_id	1	REQUEST_MEMBER revision_id	A4
				A5
R29	STUDENT student_id	1	REQUEST_MEMBER student_id	A1
				A5
R30	REQUEST_REVISION revision_id	1	RESERVATION revision_id	A4
				A5
R31	USER_ACCOUNT account_id	1	RESERVATION processed_by	A1
				A5
R32	LABORATORY lab_id	1	ITEM lab_id	A3
				A6

0..* means zero or many. A saved request needs one current revision; FK notation alone cannot enforce that.

PK-FK relationship register 3 / 4

Left cardinality = parents per child. Right = children per parent. A1-A9 identify full-definition pages.

ID	Parent / PK	Cardinality	Child / FK	Pages
R33	ITEM_CATEGORY category_id	1 ——— 0..*	ITEM category_id	A6
				A6
R34	ITEM item_id	1 ——— 0..*	STOCK_MOVEMENT item_id	A6
				A6
R35	USER_ACCOUNT account_id	1 ——— 0..*	STOCK_MOVEMENT recorded_by	A1
				A6
R36	BORROWING_ITEM borrowing_item_id	0..1 ——— 0..*	STOCK_MOVEMENT borrowing_item_id	A7
				A6
R37	RETURN_ENTRY return_entry_id	0..1 ——— 0..*	STOCK_MOVEMENT return_entry_id	A7
				A6
R38	DISPOSAL disposal_id	0..1 ——— 0..*	STOCK_MOVEMENT disposal_id	A6
				A6
R39	ITEM item_id	1 ——— 0..*	DISPOSAL item_id	A6
				A6
R40	USER_ACCOUNT account_id	1 ——— 0..*	DISPOSAL recorded_by	A1
				A6
R41	RETURN_ENTRY return_entry_id	0..1 ——— 0..*	DISPOSAL return_entry_id	A7
				A6
R42	RESERVATION reservation_id	1 ——— 0..*	BORROWING reservation_id	A5
				A7
R43	USER_ACCOUNT account_id	1 ——— 0..*	BORROWING borrower_account_id	A1
				A7
R44	USER_ACCOUNT account_id	1 ——— 0..*	BORROWING issued_by	A1
				A7
R45	BORROWING borrowing_id	1 ——— 0..*	BORROWING_ITEM borrowing_id	A7
				A7
R46	REQUEST_ITEM request_item_id	1 ——— 0..*	BORROWING_ITEM request_item_id	A5
				A7
R47	ITEM item_id	1 ——— 0..*	BORROWING_ITEM item_id	A6
				A7
R48	BORROWING borrowing_id	1 ——— 0..*	BORROWING_MEMBER borrowing_id	A7
				A7

0..* means zero or many. A saved request needs one current revision; FK notation alone cannot enforce that.

PK-FK relationship register 4 / 4

Left cardinality = parents per child. Right = children per parent. A1-A9 identify full-definition pages.

ID	Parent / PK	Cardinality	Child / FK	Pages
R49	REQUEST_MEMBER request_member_id	1 — 0..*	BORROWING_MEMBER request_member_id	A5 A7
R50	BORROWING_ITEM borrowing_item_id	1 — 0..*	RETURN_ENTRY borrowing_item_id	A7 A7
R51	USER_ACCOUNT account_id	1 — 0..*	RETURN_ENTRY received_by	A1 A7
R52	RETURN_ENTRY return_entry_id	1 — 0..*	CLEARANCE return_entry_id	A7 A8
R53	BORROWING_MEMBER borrow_member_id	1 — 0..*	CLEARANCE borrow_member_id	A7 A8
R54	USER_ACCOUNT account_id	1 — 0..*	CLEARANCE raised_by	A1 A8
R55	USER_ACCOUNT account_id	0..1 — 0..*	CLEARANCE settled_by	A1 A8
R56	RESERVATION reservation_id	1 — 0..*	USAGE_LOG reservation_id	A5 A8
R57	BORROWING borrowing_id	0..1 — 0..*	USAGE_LOG borrowing_id	A7 A8
R58	USER_ACCOUNT account_id	1 — 0..*	USAGE_LOG recorded_by	A1 A8
R59	LABORATORY lab_id	1 — 0..*	DAILY_TASK lab_id	A3 A8
R60	USER_ACCOUNT account_id	1 — 0..*	DAILY_TASK recorded_by	A1 A8
R61	USER_ACCOUNT account_id	1 — 0..*	CHAT_EXCHANGE requester_id	A1 A9

0..* means zero or many. A saved request needs one current revision; FK notation alone cannot enforce that.

Workflow rules and implementation boundary

The model describes the target persistent design; the running application still uses fixed demo accounts and temporary stores.

1. Class Representative requests

Choose Physics/Circuits, Group or Student Only, On/Out Schedule, room/time, optional equipment/materials and final review. Group accepts one or more selected students; Student Only accepts exactly one. Participant section is the authorized class section, not free-form input.

2. Faculty requests

Choose Laboratory, then Laboratory Activity or Non-Laboratory Activity. Laboratory Activity uses assigned On-Schedule room/time and skips Schedule Type. Non-Laboratory Activity chooses On/Out Schedule. Both include Schedule & Room, Equipment & Materials, and Review; Faculty has no Group/Student Only or student selection.

3. One academic reviewer, or none

Classrep On-Schedule -> assigned Faculty. Classrep Out-of-Schedule -> explicitly selected assigned Faculty or Dean. Faculty On-Schedule -> no academic approval. Faculty Non-Laboratory Activity Out-of-Schedule -> Dean. The demo has no automatic Faculty availability detection or chained approvals.

4. What submission and Dean review actually do

The server validates the form, derives requester/reference/routing, saves a detached snapshot and sets Pending Faculty/Dean Approval or Awaiting Reservation. Dean can decide only its pending Out-of-Schedule requests; rejection needs remarks. Decision history and requester status work. Faculty review is unfinished.

5. Laboratory Staff scope

Physics Staff sees Physics inventory, sample borrowing slips and submitted Physics requests. Circuits Staff sees the equivalent Circuits data. Inventory add/edit/delete works with whole-number stock and condition. Scope is server-derived from the account; querying or editing another laboratory is rejected.

6. Future reservation and borrowing connection

SERVICE_REQUEST -> current REQUEST_REVISION -> optional APPROVAL -> RESERVATION -> BORROWING -> BORROWING_ITEM / BORROWING_MEMBER -> RETURN_ENTRY -> student CLEARANCE where needed. Request approval, room booking, item issue and return are distinct records/events. Room-only usage may have no borrowing slip.

7. Database work still required

No Supabase Auth, SQL tables, RLS, live rosters, reservation holds, stock ledger, issuance/return, clearance, usage logs or assistant history are deployed. Borrowing rows and room schedules are samples. Staff stock edits do not alter the static request catalogue. Retained DFD extension tables are explicitly marked Future.

8. Integrity for integration

Validate authorized class/participants, same-lab room/item/Staff ownership, one active Classrep per section, one current revision and one final academic decision. Use transactions for live room conflicts and inventory posting. Snapshots and revision history preserve past names, quantities, approval and accountability.